

ECDAT - Cryptographic Risk Report

Scan: crypto-zoo | 2026-09-18T09:37:03 | ECDAT 0.1.0 | quantum horizon Z = 2041 (user-set) | profile CII

Executive summary

Tier	Assets	Meaning
ALREADY EXPOSED	2	Data captured today outlives the migration (Mosca $X + Y > Z$). Migration alone does not save it.
ACT NOW	39	Quantum-vulnerable or classically broken; a deadline or the Mosca margin is inside the migration window.
MONITOR	84	Vulnerable but short-lived data, signing-only, or ample time.
SAFE	72	Quantum-safe, or non-security / test-only use.

197 cryptographic assets were inventoried in CycloneDX 1.7; 62 protect data that an adversary can record today (harvest-now-decrypt-later). 41 assets must migrate first.

India (DST / National Quantum Mission, CII profile): inventory and CBOM by 2027, no new classical-only deployments after 2028, quantum-safe-only trust chains by 2029.

Migration plan

4 engineers x 6 months = 103.9 engineer-weeks; 102.1 weeks used remove 36.9% of quantum risk; 49 item(s) do not fit (0 still EXPOSED).

#	Asset	Component	Tier	Target	Effort wk	Cum. wk	Cum. risk %
1	legacy.zoo.local	pki	ACT NOW	ML-DSA-65 certificate (re-issue when the CA supports FIPS 204; hybrid/dual certificates during transition)	2.2	2.2	1.8
2	TLS	mail-capture	ACT NOW	X25519MLKEM768	2.4	4.6	3.6
3	private-key:RSA	pki	ACT NOW	-	3.6	8.2	5.4
4	RSA	legacy-hsm-agent	ALREADY EXPOSED	ML-KEM-768	8.4	16.6	9.1
5	TLS	pay-image	MONITOR	X25519MLKEM768	1.9	18.5	9.9
6	TLS	pay-image	MONITOR	X25519MLKEM768	1.9	20.4	10.7
7	public-key:RSA	pki	ACT NOW	-	4.4	24.8	12.5
8	TLS	edge-lb	MONITOR	X25519MLKEM768	1.9	26.7	13.2
9	TLS	edge-lb	MONITOR	X25519MLKEM768	1.9	28.6	13.9
10	expired.zoo.local	pay-image	MONITOR	ML-DSA-65 certificate (re-issue when the CA supports FIPS 204; hybrid/dual certificates during transition)	2.2	30.8	14.7
11	pay.zoo.local	pay-image	MONITOR	ML-DSA-65 certificate (re-issue when the CA supports FIPS 204; hybrid/dual certificates during transition)	2.2	33.0	15.6
12	MD5	legacy-hsm-agent	ACT NOW	SHA-256	8.4	41.4	18.6
13	SHA-1	legacy-hsm-agent	ACT NOW	SHA-256	8.4	49.8	21.7

#	Asset	Component	Tier	Target	Effort wk	Cum. wk	Cum. risk %
14	api.zoo.local	pki	MONITOR	ML-DSA-65 certificate (re-issue when the CA supports FIPS 204; hybrid/dual certificates during transition)	2.2	52.0	22.4
15	expired.zoo.local	pki	MONITOR	ML-DSA-65 certificate (re-issue when the CA supports FIPS 204; hybrid/dual certificates during transition)	2.2	54.2	23.1
16	jks.zoo.local	pki	MONITOR	ML-DSA-65 certificate (re-issue when the CA supports FIPS 204; hybrid/dual certificates during transition)	2.2	56.4	23.9
17	pay.zoo.local	web-capture	MONITOR	ML-DSA-65 certificate (re-issue when the CA supports FIPS 204; hybrid/dual certificates during transition)	2.2	58.6	24.6
18	ECDH	pay-image	MONITOR	X25519MLKEM768	2.7	61.3	25.4
19	api.zoo.local	pki	MONITOR	ML-DSA-65 certificate (re-issue when the CA supports FIPS 204; hybrid/dual certificates during transition)	2.4	63.7	26.1
20	TLS	web-capture	MONITOR	X25519MLKEM768	2.4	66.1	26.8
21	ECDH	edge-lb	MONITOR	X25519MLKEM768	2.4	68.5	27.5
22	RSA	edge-lb	MONITOR	ML-KEM-768	2.4	70.9	28.3
23	AES-128	legacy-hsm-agent	ACT NOW	AES-256	8.4	79.3	30.7
24	DH	pay-image	ALREADY EXPOSED	X25519MLKEM768	8.5	87.8	33.2
25	ECDSA	edge-lb	MONITOR	ML-DSA-65	2.5	90.3	33.9

Honesty notes

Z (2041) is a user-set assumption anchored to the Global Risk Institute 2025 survey (CRQC 28-49% within 10 years, 51-70% within 15). Mosca's inequality is applied only to confidentiality primitives; signatures and hashes are deadline-driven. Binary findings are best-effort. Every finding carries confidence and file:line evidence. Patches are suggestions and are never applied automatically.

Technical appendix: all assets

Tier	Asset	Type	Component	Where	Class	X/Y/Z	Agility	Recommendation	Conf.
ALREADY EXPOSED	RSA	algorithm	legacy-hsm-agent	legacy.c:9	broken HNDL	30/3.8/15	30	ML-KEM-768	high
ACT NOW	MD5	algorithm	legacy-hsm-agent	legacy.c:18	legacy-broken	30/3.8/15	30	SHA-256	high
ACT NOW	SHA-1	algorithm	legacy-hsm-agent	legacy.c:21	legacy-broken	30/3.8/15	30	SHA-256	high
ACT NOW	AES-128	algorithm	legacy-hsm-agent	legacy.c:13	weakened HNDL	30/3.8/15	30	AES-256	high
ALREADY EXPOSED	DH	algorithm	pay-image	image://pay-image/etc/ssl/openssl.cnf	broken HNDL	10/7.76/15	28	X25519MLKEM768	medium
ACT NOW	RSA	algorithm	payments-api	payments/auth.py:12	broken HNDL	10/3.8/15	30	ML-KEM-768	high
ACT NOW	ECDSA	algorithm	payments-api	payments/auth.py:20	broken	10/3.8/15	30	ML-DSA-65	high
ACT NOW	MD5	algorithm	core-banking	PaymentCrypto.java:14	legacy-broken	10/3.72/15	32	SHA-256	high
ACT NOW	3DES	algorithm	core-banking	PaymentCrypto.java:15	legacy-broken	10/3.72/15	32	AES-256-GCM	high
ACT NOW	SHA-1	algorithm	pay-image	image://pay-image/etc/ssl/openssl.cnf	legacy-broken	10/8.32/15	21	SHA-256	high
ACT NOW	MD5	algorithm	pay-image	image://pay-image/usr/lib/libcrypto.so.1.1	legacy-broken	10/8.16/15	23	SHA-256	medium
ACT NOW	MD5	algorithm	web-portal	server.js:6	legacy-broken	7/3.8/15	30	SHA-256	high
ACT NOW	RSA	algorithm	web-portal	server.js:15	legacy-broken	7/3.76/15	31	ML-KEM-768	high
ACT NOW	SHA-1	algorithm	edge-lb	openssl.cnf	legacy-broken	5/7.76/15	28	SHA-256	medium
ACT NOW	private-key:RSA	related-crypto-material	pki	rsa1024.key:1	legacy-broken	5/1/15	70	-	high
ACT NOW	RSA	algorithm	pki	rsa1024_sha1.der	legacy-broken	5/3.48/15	38	ML-KEM-768	high
ACT NOW	legacy.zoo.local	certificate	pki	rsa1024_sha1.der	legacy-broken	5/0.5/15	76	ML-DSA-65 certificate (re-issue when the CA supports FIPS 204; hybrid/dual certificates during transition)	high
ACT NOW	SHA-1	algorithm	pki	rsa1024_sha1.der	legacy-broken	5/3.4/15	40	SHA-256	high
ACT NOW	public-key:RSA	related-crypto-material	pki	rsa1024_sha1.der	legacy-broken	5/1/15	58	-	medium
ACT NOW	TLS	protocol	mail-capture	pcap://smtp_starttls.pcap#10.0.0.25:25	legacy-broken	5/0.5/15	70	X25519MLKEM768	high
ACT NOW	3DES	algorithm	mail-capture	pcap://smtp_starttls.pcap#10.0.0.25:25	legacy-broken	5/3.6/15	35	AES-256-GCM	high
ACT NOW	RSA	algorithm	core-banking	PaymentCrypto.java:9	broken HNDL	10/3.72/15	32	ML-KEM-768	high
ACT NOW	ECDSA	algorithm	core-banking	PaymentCrypto.java:13	broken	10/3.72/15	32	ML-DSA-65	high
ACT NOW	RSA	algorithm	pay-image	image://pay-image/etc/ssl/certs/legacy.der	broken HNDL	10/4.44/15	14	ML-KEM-768	high
ACT NOW	AES-128	algorithm	pay-image	image://pay-image/etc/ssl/openssl.cnf	weakened HNDL	10/7.76/15	28	AES-256	medium

Tier	Asset	Type	Component	Where	Class	X/Y/Z	Agility	Recommendation	Conf.
ACT NOW	AES	algorithm	pay-image	image://pay-image/usr/lib/libcrypto.so.1.1	weakened HNDL	10/8.16/15	23	AES-256	medium
ACT NOW	RSA	algorithm	web-portal	server.js:14	broken HNDL	7/3.8/15	30	ML-KEM-768	high
ACT NOW	ECDSA	algorithm	gateway	main.go:18	broken	5/3.48/15	38	ML-DSA-65	high
ACT NOW	RSA	algorithm	gateway	main.go:17	broken HNDL	5/3.48/15	38	ML-KEM-768	high
ACT NOW	DH	algorithm	edge-lb	openssl.cnf	broken HNDL	5/7.76/15	28	X25519MLKEM768	medium
ACT NOW	RSA	algorithm	pki	keystore.jks	broken HNDL	5/3.92/15	27	ML-KEM-768	high
ACT NOW	RSA	algorithm	pki	keystore.jks	broken HNDL	5/3.68/15	33	ML-KEM-768	high
ACT NOW	ECDSA	algorithm	pki	modern.jks	broken	5/3.52/15	37	ML-DSA-65	high
ACT NOW	RSA	algorithm	mail-capture	pcap://smtp_starttls.pcap#10.0.0.25:25	broken HNDL	5/3.6/15	35	ML-KEM-768	high
ACT NOW	ECDH	algorithm	web-capture	pcap://tls_handshake.pcap#10.0.0.80:443	broken HNDL	5/3.6/15	35	X25519MLKEM768	high
ACT NOW	RSA	algorithm	web-capture	pcap://tls_handshake.pcap#10.0.0.80:443	broken HNDL	5/3.6/15	35	ML-KEM-768	high
ACT NOW	MD5	algorithm	zoobin	libzooecrypto.so	legacy-broken	5/8/15	25	SHA-256	medium
ACT NOW	SHA-1	algorithm	zoobin	libzooecrypto.so	legacy-broken	5/8/15	25	SHA-256	medium
ACT NOW	RSA	algorithm	zoobin	libzooecrypto.so	broken HNDL	5/8/15	25	ML-KEM-768	medium
ACT NOW	ECDH	algorithm	zoobin	zoobin	broken HNDL	5/7.76/15	28	X25519MLKEM768	medium
ACT NOW	ECDSA	algorithm	zoobin	zoobin	broken	5/7.76/15	28	ML-DSA-65	medium
MONITOR	AES	algorithm	core-banking	PaymentCrypto.java:8	weakened HNDL	10/3.48/15	38	AES-256	high
MONITOR	ECDSA	algorithm	pay-image	image://pay-image/etc/nginx/nginx.conf:8	broken	10/0.5/15	66	ML-DSA-65	high
MONITOR	TLS	protocol	pay-image	image://pay-image/etc/nginx/nginx.conf:7	broken HNDL	10/0.5/15	81	X25519MLKEM768	high
MONITOR	ECDH	algorithm	pay-image	image://pay-image/etc/nginx/nginx.conf:8	broken HNDL	10/0.5/15	68	X25519MLKEM768	high
MONITOR	TLS	protocol	pay-image	image://pay-image/etc/ssl/openssl.cnf:11	broken HNDL	10/0.5/15	81	X25519MLKEM768	high
MONITOR	expired.zoo.local	certificate	pay-image	image://pay-image/etc/ssl/certs/legacy.der	broken HNDL	10/0.5/15	76	ML-DSA-65 certificate (re-issue when the CA supports FIPS 204; hybrid/dual certificates during transition)	high
MONITOR	pay.zoo.local	certificate	pay-image	image://pay-image/etc/ssl/certs/pay.crt:1	broken HNDL	10/0.5/15	76	ML-DSA-65 certificate (re-issue when the CA supports FIPS 204; hybrid/dual certificates during transition)	high

Tier	Asset	Type	Component	Where	Class	X/Y/Z	Agility	Recommendation	Conf.
MONITOR	public-key:RSA	related-crypto-material	pay-image	image://pay-image/etc/ssl/certs/legacy.der	broken HNDL	10/1/15	58	-	medium
MONITOR	public-key:RSA	related-crypto-material	pay-image	image://pay-image/etc/ssl/certs/pay.crt	broken HNDL	10/1/15	58	-	medium
MONITOR	AES	algorithm	gateway	main.go:20	weakened HNDL	5/3.48/15	38	AES-256	high
MONITOR	TLS	protocol	gateway	main.go:22	broken HNDL	5/0.5/15	67	X25519MLKEM768	medium
MONITOR	TLS	protocol	edge-lb	nginx.conf:7	broken HNDL	5/0.5/15	81	X25519MLKEM768	high
MONITOR	ECDH	algorithm	edge-lb	nginx.conf:8	broken HNDL	5/0.5/15	74	X25519MLKEM768	high
MONITOR	RSA	algorithm	edge-lb	nginx.conf:8	broken HNDL	5/0.5/15	74	ML-KEM-768	high
MONITOR	ECDSA	algorithm	edge-lb	nginx.conf:8	broken	5/0.5/15	73	ML-DSA-65	high
MONITOR	TLS	protocol	edge-lb	openssl.cnf:11	broken HNDL	5/0.5/15	81	X25519MLKEM768	high
MONITOR	AES-128	algorithm	edge-lb	openssl.cnf	weakened HNDL	5/7.76/15	28	AES-256	medium
MONITOR	pay.zoo.local	certificate	pki	pay.p12	broken HNDL	5/0.5/15	70	ML-DSA-65 certificate (re-issue when the CA supports FIPS 204; hybrid/dual certificates during transition)	high
MONITOR	ECDAT Zoo Root CA	certificate	pki	keystore.jks	broken HNDL	5/0.5/15	68	ML-DSA-65 certificate (re-issue when the CA supports FIPS 204; hybrid/dual certificates during transition)	high
MONITOR	api.zoo.local	certificate	pki	ecdsa_p256.crt:1	broken	5/0.5/15	76	ML-DSA-65 certificate (re-issue when the CA supports FIPS 204; hybrid/dual certificates during transition)	high
MONITOR	private-key:RSA	related-crypto-material	pki	keystore.jks	broken HNDL	5/1/15	70	-	high
MONITOR	jks.zoo.local	certificate	pki	keystore.jks	broken HNDL	5/0.5/15	76	ML-DSA-65 certificate (re-issue when the CA supports FIPS 204; hybrid/dual certificates during transition)	high

Tier	Asset	Type	Component	Where	Class	X/Y/Z	Agility	Recommendation	Conf.
MONITOR	api.zoo.local	certificate	pki	modern.jks	broken	5/0.5/15	74	ML-DSA-65 certificate (re-issue when the CA supports FIPS 204; hybrid/dual certificates during transition)	high
MONITOR	private-key:ECDSA	related-crypto-material	pki	modern.jks	broken	5/1/15	70	-	high
MONITOR	private-key:RSA	related-crypto-material	pki	pay.p12	broken HNDL	5/1/15	70	-	high
MONITOR	expired.zoo.local	certificate	pki	rsa2048_expired.crt	broken HNDL	5/0.5/15	76	ML-DSA-65 certificate (re-issue when the CA supports FIPS 204; hybrid/dual certificates during transition)	high
MONITOR	private-key:RSA	related-crypto-material	pki	rsa2048_sha256.key:1	broken HNDL	5/1/15	70	-	high
MONITOR	public-key:RSA	related-crypto-material	pki	bundle.pem	broken HNDL	5/1/15	58	-	medium
MONITOR	public-key:RSA	related-crypto-material	pki	ca_rsa4096.crt	broken HNDL	5/1/15	58	-	medium
MONITOR	public-key:ECDSA	related-crypto-material	pki	ecdsa_p256.crt	broken	5/1/15	58	-	medium
MONITOR	public-key:RSA	related-crypto-material	pki	rsa2048_expired.crt	broken HNDL	5/1/15	58	-	medium
MONITOR	public-key:RSA	related-crypto-material	pki	rsa2048_sha256.crt	broken HNDL	5/1/15	58	-	medium
MONITOR	unknown:pkcs12-file	related-crypto-material	pki	pay.p12	unknown	5/1/15	58	-	medium
MONITOR	TLS	protocol	web-capture	pcap://tls_handshake.pcap#10.0.0.80:443	broken HNDL	5/0.5/15	70	X25519MLKEM768	high
MONITOR	pay.zoo.local	certificate	web-capture	pcap://tls_handshake.pcap#10.0.0.80:443	broken HNDL	5/0.5/15	76	ML-DSA-65 certificate (re-issue when the CA supports FIPS 204; hybrid/dual certificates during transition)	high
MONITOR	AES	algorithm	zoobin	libzoocrypto.so	weakened HNDL	5/8/15	25	AES-256	medium
MONITOR	TLS	protocol	zoobin	zoobin	broken HNDL	5/1/15	57	X25519MLKEM768	medium
SAFE	SHA-256	algorithm	payments-api	utils/cache.py:10	safe	10/3.88/15	28	-	high
SAFE	AES-256	algorithm	payments-api	payments/auth.py:25	safe	10/3.6/15	35	-	high

Tier	Asset	Type	Component	Where	Class	X/Y/Z	Agility	Recommendation	Conf.
SAFE	SHA-1	algorithm	payments-api	tests/test_auth.py:6	legacy-broken	10/3.76/15	31	-	high
SAFE	MD5	algorithm	payments-api	utils/cache.py:6	legacy-broken	10/3.76/15	31	-	high
SAFE	SHA-256	algorithm	gateway	main.go:19	safe	5/3.32/15	42	-	high
SAFE	SHA-256	algorithm	web-portal	server.js:1	safe	7/3.64/15	34	-	low
SAFE	AES-256	algorithm	web-portal	server.js:10	safe	7/3.8/15	30	-	high
SAFE	HMAC	algorithm	web-portal	server.js:16	safe	7/3.6/15	35	-	high
SAFE	AES-256	algorithm	edge-lb	nginx.conf:8	safe	5/0.5/15	74	-	high
SAFE	SHA-384	algorithm	edge-lb	nginx.conf:8	safe	5/0.5/15	74	-	high
SAFE	ChaCha20	algorithm	edge-lb	nginx.conf:8	safe	5/0.5/15	74	-	high
SAFE	SHA-256	algorithm	edge-lb	openssl.cnf	safe	5/7.76/15	28	-	medium
SAFE	SHA-256	algorithm	pki	keystore.jks	safe	5/3.92/15	27	-	high
SAFE	SHA-384	algorithm	pki	ecdsa_p256.crt:1	safe	5/3.44/15	39	-	high
SAFE	SHA-256	algorithm	zoobin	libzocrypto.so	safe	5/8/15	25	-	medium
SAFE	SHA-512	algorithm	zoobin	zoobin	safe	5/7.76/15	28	-	medium
SAFE	SHA3-256	algorithm	zoobin	zoobin	safe	5/7.76/15	28	-	medium
SAFE	ChaCha20	algorithm	zoobin	zoobin	safe	5/7.76/15	28	-	medium
SAFE	SHA-256	algorithm	pay-image	image://pay-image/etc/ssl/certs/legacy.der	safe	10/4.44/15	14	-	high
SAFE	AES-256	algorithm	pay-image	image://pay-image/app/payments/auth.py:25	safe	10/3.76/15	31	-	high
SAFE	SHA-384	algorithm	pay-image	image://pay-image/etc/nginx/nginx.conf:8	safe	10/0.5/15	68	-	high
SAFE	ChaCha20	algorithm	pay-image	image://pay-image/etc/nginx/nginx.conf:8	safe	10/0.5/15	68	-	high
SAFE	HMAC	algorithm	mail-capture	pcap://smtp_starttls.pcap#10.0.0.25:25	safe	5/3.6/15	35	-	high
SAFE	AES-256	algorithm	web-capture	pcap://tls_handshake.pcap#10.0.0.80:443	safe	5/3.6/15	35	-	high
SAFE	SHA-384	algorithm	web-capture	pcap://tls_handshake.pcap#10.0.0.80:443	safe	5/3.6/15	35	-	high
SAFE	SHA-256	algorithm	web-capture	pcap://tls_handshake.pcap#10.0.0.80:443	safe	5/3.4/15	40	-	high
MONITOR	ECDH	algorithm	payments-api	requirements.txt	broken HNDL	10/3.64/15	34	X25519MLKEM768	low
MONITOR	X25519	algorithm	payments-api	requirements.txt	broken HNDL	10/3.64/15	34	X25519MLKEM768	low
MONITOR	Ed25519	algorithm	payments-api	requirements.txt	broken	10/3.64/15	34	ML-DSA-65	low
MONITOR	DH	algorithm	payments-api	requirements.txt	broken HNDL	10/3.64/15	34	X25519MLKEM768	low
MONITOR	DSA	algorithm	payments-api	requirements.txt	broken	10/3.64/15	34	ML-DSA-65	low
MONITOR	AES-128	algorithm	payments-api	requirements.txt	weakened HNDL	10/3.64/15	34	AES-256	low
SAFE	ChaCha20	algorithm	payments-api	requirements.txt	safe	10/3.64/15	34	-	low

Tier	Asset	Type	Component	Where	Class	X/Y/Z	Agility	Recommendation	Conf.
MONITOR	3DES	algorithm	payments-api	requirements.txt	legacy-broken	10/3.64/15	34	AES-256-GCM	low
SAFE	SHA-384	algorithm	payments-api	requirements.txt	safe	10/3.64/15	34	-	low
SAFE	SHA-512	algorithm	payments-api	requirements.txt	safe	10/3.64/15	34	-	low
SAFE	SHA3-256	algorithm	payments-api	requirements.txt	safe	10/3.64/15	34	-	low
SAFE	HMAC	algorithm	payments-api	requirements.txt	safe	10/3.64/15	34	-	low
SAFE	PBKDF2	algorithm	payments-api	requirements.txt	safe	10/3.64/15	34	-	low
SAFE	scrypt	algorithm	payments-api	requirements.txt	safe	10/3.64/15	34	-	low
SAFE	HKDF	algorithm	payments-api	requirements.txt	safe	10/3.64/15	34	-	low
MONITOR	DES	algorithm	payments-api	requirements.txt	legacy-broken	10/3.64/15	34	AES-256-GCM	low
MONITOR	RC4	algorithm	payments-api	requirements.txt	legacy-broken	10/3.64/15	34	AES-256-GCM	low
MONITOR	Blowfish	algorithm	payments-api	requirements.txt	legacy-broken	10/3.64/15	34	AES-256-GCM	low
MONITOR	ECDH	algorithm	core-banking	pom.xml	broken HNDL	10/3.64/15	34	X25519MLKEM768	low
MONITOR	X25519	algorithm	core-banking	pom.xml	broken HNDL	10/3.64/15	34	X25519MLKEM768	low
MONITOR	Ed25519	algorithm	core-banking	pom.xml	broken	10/3.64/15	34	ML-DSA-65	low
MONITOR	DH	algorithm	core-banking	pom.xml	broken HNDL	10/3.64/15	34	X25519MLKEM768	low
MONITOR	DSA	algorithm	core-banking	pom.xml	broken	10/3.64/15	34	ML-DSA-65	low
MONITOR	AES-128	algorithm	core-banking	pom.xml	weakened HNDL	10/3.64/15	34	AES-256	low
SAFE	AES-256	algorithm	core-banking	pom.xml	safe	10/3.64/15	34	-	low
SAFE	ChaCha20	algorithm	core-banking	pom.xml	safe	10/3.64/15	34	-	low
MONITOR	DES	algorithm	core-banking	pom.xml	legacy-broken	10/3.64/15	34	AES-256-GCM	low
MONITOR	RC4	algorithm	core-banking	pom.xml	legacy-broken	10/3.64/15	34	AES-256-GCM	low
MONITOR	Blowfish	algorithm	core-banking	pom.xml	legacy-broken	10/3.64/15	34	AES-256-GCM	low
MONITOR	SHA-1	algorithm	core-banking	pom.xml	legacy-broken	10/3.64/15	34	SHA-256	low
SAFE	SHA-256	algorithm	core-banking	pom.xml	safe	10/3.64/15	34	-	low
SAFE	SHA-384	algorithm	core-banking	pom.xml	safe	10/3.64/15	34	-	low
SAFE	SHA-512	algorithm	core-banking	pom.xml	safe	10/3.64/15	34	-	low
SAFE	SHA3-256	algorithm	core-banking	pom.xml	safe	10/3.64/15	34	-	low
SAFE	HMAC	algorithm	core-banking	pom.xml	safe	10/3.64/15	34	-	low
SAFE	PBKDF2	algorithm	core-banking	pom.xml	safe	10/3.64/15	34	-	low

Tier	Asset	Type	Component	Where	Class	X/Y/Z	Agility	Recommendation	Conf.
SAFE	scrypt	algorithm	core-banking	pom.xml	safe	10/3.64/15	34	-	low
SAFE	Argon2	algorithm	core-banking	pom.xml	safe	10/3.52/15	37	-	low
SAFE	ML-KEM-768	algorithm	core-banking	pom.xml	safe	10/3.64/15	34	-	low
SAFE	ML-KEM-1024	algorithm	core-banking	pom.xml	safe	10/3.64/15	34	-	low
SAFE	ML-DSA-65	algorithm	core-banking	pom.xml	safe	10/3.64/15	34	-	low
SAFE	ML-DSA-87	algorithm	core-banking	pom.xml	safe	10/3.64/15	34	-	low
SAFE	SLH-DSA	algorithm	core-banking	pom.xml	safe	10/3.64/15	34	-	low
SAFE	FN-DSA	algorithm	core-banking	pom.xml	safe	10/3.64/15	34	-	low
SAFE	ChaCha20	algorithm	gateway	go.mod	safe	5/3.64/15	34	-	low
MONITOR	Ed25519	algorithm	gateway	go.mod	broken	5/3.64/15	34	ML-DSA-65	low
MONITOR	X25519	algorithm	gateway	go.mod	broken HNDL	5/3.64/15	34	X25519MLK EM768	low
SAFE	bcrypt	algorithm	gateway	go.mod	safe	5/3.52/15	37	-	low
SAFE	scrypt	algorithm	gateway	go.mod	safe	5/3.64/15	34	-	low
SAFE	Argon2	algorithm	gateway	go.mod	safe	5/3.52/15	37	-	low
SAFE	PBKDF2	algorithm	gateway	go.mod	safe	5/3.64/15	34	-	low
SAFE	HKDF	algorithm	gateway	go.mod	safe	5/3.64/15	34	-	low
SAFE	SHA3-256	algorithm	gateway	go.mod	safe	5/3.64/15	34	-	low
MONITOR	Blowfish	algorithm	gateway	go.mod	legacy-broken	5/3.64/15	34	AES-256-GCM	low
MONITOR	SSH	algorithm	gateway	go.mod	unknown	5/3.64/15	34	-	low
MONITOR	AES-128	algorithm	web-portal	package.json	weakened HNDL	7/3.64/15	34	AES-256	low
MONITOR	3DES	algorithm	web-portal	package.json	legacy-broken	7/3.64/15	34	AES-256-GCM	low
MONITOR	DES	algorithm	web-portal	package.json	legacy-broken	7/3.64/15	34	AES-256-GCM	low
MONITOR	RC4	algorithm	web-portal	package.json	legacy-broken	7/3.64/15	34	AES-256-GCM	low
MONITOR	SHA-1	algorithm	web-portal	package.json	legacy-broken	7/3.64/15	34	SHA-256	low
SAFE	SHA-384	algorithm	web-portal	package.json	safe	7/3.64/15	34	-	low
SAFE	SHA-512	algorithm	web-portal	package.json	safe	7/3.64/15	34	-	low
SAFE	PBKDF2	algorithm	web-portal	package.json	safe	7/3.64/15	34	-	low
MONITOR	TLS	algorithm	web-portal	package.json	unknown	7/3.64/15	34	-	low
SAFE	SHA3-256	algorithm	web-portal	package.json	safe	7/3.64/15	34	-	low

Tier	Asset	Type	Component	Where	Class	X/Y/Z	Agility	Recommendation	Conf.
SAFE	bcrypt	algorithm	web-portal	package.json	safe	7/3.52/15	37	-	low
MONITOR	X25519	algorithm	zoobin	libzoocrypto.so	broken HNDL	5/3.64/15	34	X25519MLKEM768	low
MONITOR	Ed25519	algorithm	zoobin	libzoocrypto.so	broken	5/3.64/15	34	ML-DSA-65	low
MONITOR	DH	algorithm	zoobin	libzoocrypto.so	broken HNDL	5/3.64/15	34	X25519MLKEM768	low
MONITOR	DSA	algorithm	zoobin	libzoocrypto.so	broken	5/3.64/15	34	ML-DSA-65	low
MONITOR	AES-128	algorithm	zoobin	libzoocrypto.so	weakened HNDL	5/3.64/15	34	AES-256	low
SAFE	AES-256	algorithm	zoobin	libzoocrypto.so	safe	5/3.64/15	34	-	low
MONITOR	3DES	algorithm	zoobin	libzoocrypto.so	legacy-broken	5/3.64/15	34	AES-256-GCM	low
MONITOR	DES	algorithm	zoobin	libzoocrypto.so	legacy-broken	5/3.64/15	34	AES-256-GCM	low
MONITOR	RC4	algorithm	zoobin	libzoocrypto.so	legacy-broken	5/3.64/15	34	AES-256-GCM	low
SAFE	SHA-384	algorithm	zoobin	libzoocrypto.so	safe	5/3.64/15	34	-	low
SAFE	HMAC	algorithm	zoobin	libzoocrypto.so	safe	5/3.64/15	34	-	low
SAFE	PBKDF2	algorithm	zoobin	libzoocrypto.so	safe	5/3.64/15	34	-	low
MONITOR	TLS	algorithm	zoobin	libzoocrypto.so	unknown	5/3.64/15	34	-	low
MONITOR	X25519	algorithm	pay-image	image://pay-image/app/requirements.txt	broken HNDL	10/3.64/15	34	X25519MLKEM768	low
MONITOR	Ed25519	algorithm	pay-image	image://pay-image/app/requirements.txt	broken	10/3.64/15	34	ML-DSA-65	low
MONITOR	DSA	algorithm	pay-image	image://pay-image/app/requirements.txt	broken	10/3.64/15	34	ML-DSA-65	low
MONITOR	3DES	algorithm	pay-image	image://pay-image/app/requirements.txt	legacy-broken	10/3.64/15	34	AES-256-GCM	low
SAFE	SHA-512	algorithm	pay-image	image://pay-image/app/requirements.txt	safe	10/3.64/15	34	-	low
SAFE	SHA3-256	algorithm	pay-image	image://pay-image/app/requirements.txt	safe	10/3.64/15	34	-	low
SAFE	HMAC	algorithm	pay-image	image://pay-image/app/requirements.txt	safe	10/3.64/15	34	-	low
SAFE	PBKDF2	algorithm	pay-image	image://pay-image/app/requirements.txt	safe	10/3.64/15	34	-	low
SAFE	scrypt	algorithm	pay-image	image://pay-image/app/requirements.txt	safe	10/3.64/15	34	-	low
SAFE	HKDF	algorithm	pay-image	image://pay-image/app/requirements.txt	safe	10/3.64/15	34	-	low
MONITOR	DES	algorithm	pay-image	image://pay-image/app/requirements.txt	legacy-broken	10/3.64/15	34	AES-256-GCM	low
MONITOR	RC4	algorithm	pay-image	image://pay-image/app/requirements.txt	legacy-broken	10/3.64/15	34	AES-256-GCM	low
MONITOR	Blowfish	algorithm	pay-image	image://pay-image/app/requirements.txt	legacy-broken	10/3.64/15	34	AES-256-GCM	low
MONITOR	TLS	algorithm	pay-image	image://pay-image/usr/lib/libcrypto.so.1.1	unknown	10/3.64/15	34	-	low